

EMBEDDED SYSTEM AND DESIGN

EXPERIMENT NO. 2

NUMBER SYSTEM EMULATOR

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DISCUSSION

Number system is defined as a system of naming or representing numbers associated with their chosen base system. In mathematics, there are several number systems such as the decimal, binary, octal, and hexadecimal number system. Conversion of these number systems are essential in the relationship between machine languages and human languages. With that being said, a number system emulator was recreated through the use of the arduino kit, particularly the arduino uno, LEDs, buttons, infrared sensor and remote, and seven-segment display.

There are two parts of the experiment: Decimal to 4-bit binary and 4-bit binary to its excess-5 in decimal form. In the first part, an infrared sensor was utilized to receive the signal from the IR remote that sends signals that are in hexadecimal form. The signal will then be assimilated into a switch case, which determines

what number was pressed. Subsequently, the said number will be exhibited in the 7-segment display. Note that numbers on the remote only include 0-9, hence only those numbers could be the input for the first conversion. The decimal that was now displayed on the seven-segment display would also be converted and shown on the LEDs in a binary form of 4-bits. These processes are possible through the use of two external libraries: *SevSeg.h* for the 4-digit seven-segment display, and *IRremote.h* for the infrared receiver and remote. As per usual, the *analogWrite* was utilized to turn the LEDs on or off with the use of a switch case that determines the number.

The second part involves a conversion from 4-bit binary to excess-5 but in decimal form. This approach employs switch buttons, LEDs, and the 7-segment. The process is that the inputs are the switch buttons. LOW will be a button's value if pressed, while unpressed buttons are HIGH. So, the 4 switch buttons serve as the 4-bits in a binary

value. There will be a three-count blink to give time for the user in pressing the buttons. After the three counts had passed, the decimal-form excess-5 of the binary input will be shown in the seven-segment display; hence, converting the binary to its excess-5 form. The process simply involves a switch case, if else, and analogWrite to turn the LEDs on.

CONCLUSION

In conclusion, the number systems conversion emulator was demonstrated with the use of the arduino kit, converting decimal to 4-bit binary and 4-bit binary to decimal-for excess-5. For the application, the LEDs were for the 4-bit binary value, the switch buttons were the input for the binary value, the seven-segment display was used for the decimal value, and the infrared sensor and remote were employed for the decimal value. The lessons in microprocessors, logic circuits, and switching theory were applied by configuring the status of both the LEDs and seven-segment display, applying the circuit's function and conversion, and connecting wires to its proper pins. Therefore, the number systems conversion emulator, an embedded system, was successfully and satisfactorily produced. All objectives were met for this experiment.